



KARUNA VASU

SOFTWARE ENGINEER

CONTACT

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[Email](#) | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C#, PERL, R

Backend & APIs: FastAPI, REST APIs, microservices, ASP.NET Core (.NET 9), Hangfire, Flask, Django, SQLAlchemy, JWT RBAC

Frontend: React, Next.js, Vite, Tailwind CSS

Automation: Selenium, RPA, workflow automation, Aavailability RCM pipelines

AI/ML: Azure OpenAI, Amazon Bedrock (Claude 3.5), OCR (RapidOCR), LLMs, prompt engineering, document AI

Data Processing: Pandas, NumPy, rule engines, data validation, openpyxl

Cloud & DevOps: AWS (Connect, Bedrock, Lambda, EC2, DynamoDB, IAM), Docker, CI/CD, Azure Key Vault, Azure DevOps

Databases: PostgreSQL, Microsoft SQL Server, SQL, SQLite, vector DB (basic)

Platform & Tooling: APScheduler, Serilog / structured logging, PyODBC, SFTP automation, Git, Linux, Postman, Jenkins, Kubernetes, Power Automate

EDUCATION

2021 - 2023

Master of Science: Bioinformatics
Manipal Academy of Higher Education,
Manipal, India
GPA: 7.24

2018 - 2021

Bachelor of Science: Bioinformatics
Sardar Patel University Gujarat, Anand, India
GPA: 7.00

PROFESSIONAL SUMMARY

Software Engineer building automation and AI systems for US Healthcare RCM. Delivered ~95% faster claim processing and 40–80% less manual work.

Core work spans App Hub / BotVeta orchestration, Automation Productivity Dashboard (tracking 371 projects, ~118K jobs/week, ~\$20M charge-cost visibility), Azure OpenAI PDF extraction, Slack SLA operations, and AWS IVR automation.

EXPERIENCE

December 18, 2023 - Present

Software Engineer, Integrity Healthcare Solutions Pvt Ltd, Ahmedabad

- Reduced claim turnaround by **~95%** with Python automation pipelines integrated with **Aavailability** and RCM workflows.
- Automated eligibility, authorization, claims, and EOB/ERA flows using Python and Selenium; cut manual effort by **40–80%**.
- Built **FastAPI** microservices for high-volume production automation with retries and SQL Server logging.
- Architected **BotVeta / App Hub**: Hangfire schedules, Audit Center, auth rate limits on login/OTP, product-usage analytics, and **Product Weekly Update** (UTC date ranges + comparison).
- Built **Automation Productivity Dashboard** tracking 371 ADO projects (PM/SR CSM, discussions + status/color), **~118K** jobs/week (**84%** success), and 2,770 FTE sent (**~\$20M charge-cost visibility**).
- Shipped **AI PDF Extractor** (Azure OpenAI + OCR): Appointment List / HCO tracker / Trust-PHNT-NBA templates with validated **saved_minutes** ✕ FTE KPIs.
- Delivered **Slack Thread Tracker**: auto-resolve image-only messages, reaction-driven status, pre-shift digests, AI triage, and SLA analytics.
- Engineered IVR automation on Amazon Connect + AWS Bedrock for payer calls; reduced manual calling effort by **60–80%**.

June 2023 - July 2023

Data Science Intern, Zephyr Technologies and Solutions Pvt Ltd, Mangalore, India

- Developed a real-time chat application using **Django, WebSockets, and PostgreSQL**.

February 2023 - December 2023

Research Intern, Manipal Institute of Technology, Manipal, India

- Published IEEE research paper on ECG signal analysis; awarded **Best Paper & Best Presenter** (NMITCON 2023).

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